



Bayesian decision theory

S : state of the world (unknown)

a : possible action

y : obs. data

$p(s|y)$: post. dist. over states given data

$L(s,a)$: the loss incurred when the true state is s and you take action a .

$E_{p(s|y)}[L(s,a)]$ is a post. exp. loss: tells us the average loss we'd expect if we took action a given what you know (y).

$L(s,a)$	Do nothing	Isolate	Hospitalize
Healthy	0	10	50
C-19 mild	10	0	20
C-19 severe	100	10	0

Spec. doctor arrived at following posterior predictions for when pat. gets covid or not

State s	s_1 Healthy	s_2 Mild symp.	s_3 Severe
$p(s y)$	0.65	0.3	0.05

- we use:
$$E_{p(s|y)}[L(s,a)] = \sum_s p(s|y) \cdot L(s,a)$$

$$A_1: E[L(s | \text{Do nothing})] = 0.65 \cdot 0 + 0.3 \cdot 10 + 0.05 \cdot 100 = \underline{8}$$

$$A_2: E[L(s | \text{isolate at h})] = 0.65 \cdot 10 + 0.3 \cdot 0 + 0.05 \cdot 10 = 7$$

$$A_3: E[L(s | \text{hosp.})] = 0.65 \cdot 50 + 0.3 \cdot 20 + 0.05 \cdot 0 = 38.5$$

$\therefore A_2$ minimizes the loss

From Bernoulli to Softmax

- $\forall$ data pts n , we compute vector of logits (scores) $f_n \in \mathbb{R}^k$
- we turn these logits (scores) into class probabilities w/ softmax:

$$\hat{\pi}_{n,i} = \text{softmax}(f_n)_i = \frac{\exp(f_{n,i})}{\sum_{j=1}^k \exp(f_{n,j})}, \quad i=1, \dots, k$$

- then we say the label y_n is a categorical random variable w/ these probabilities: $y_n | f_n \sim \text{Cat}[\text{softmax}(f_n)]$, i.e. $P(y_n = i | f_n) = \hat{\pi}_{n,i}$

example: from logits to probs.

$k=3$ and we get the following logits, $f_n = [2, 0, -1]$

we then exponentiate: $\left. \begin{array}{l} \exp(f_{n,1}) = 7.39 \\ \exp(f_{n,2}) = 1.00 \\ \exp(f_{n,3}) = 0.37 \end{array} \right\} \circ \text{ we add these } \Rightarrow \sum_{j=1}^3 \exp(f_{n,j}) = \exp(2) + \exp(0) + \exp(-1)$

$$= 7.39 + 1.00 + 0.37 = 8.76$$

or writing more compactly as

$$\hat{\pi}_{n,1} = \frac{\exp(f_{n,1})}{\sum_{j=1}^3 \exp(f_{n,j})} = \frac{e(2)}{e(2) + e(0) + e(-1)} \approx \frac{7.39}{7.39 + 1.00 + 0.37} \approx \frac{7.39}{8.76} \approx 0.843$$

after we normalize all 3 entries, we obtain: $\hat{\pi}_n = \left[\frac{7.39}{8.76}, \frac{1.00}{8.76}, \frac{0.37}{8.76} \right] \approx [0.843, 0.114, 0.042]$

$\hookrightarrow$ given these scores, the model assigns about 84% prob. to class 1, 10% to class 2 and 4% to class 3

if you sample y_n from $\text{Categorical}(\hat{\pi}_n)$, you'll draw class 1 most of a time

if you predict, we would take $\arg \max_i \hat{\pi}_{n,i} \Rightarrow \text{class 1}$

example w/ linear model

input $x_n = [1, 0, 2]^T$ and use linear map: $f_n = W^T x_n + b$, $W = \begin{bmatrix} 1 & 0 & -1 \\ 0.5 & -0.5 & 0 \end{bmatrix}$, $b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

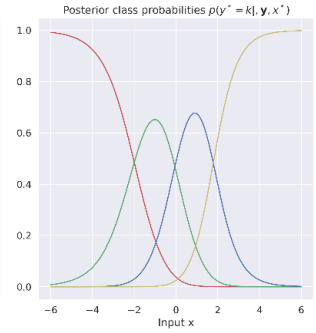
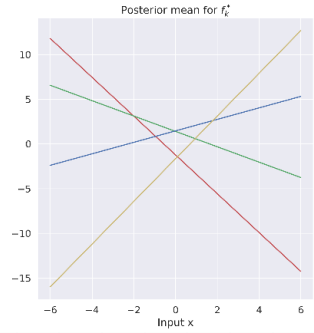
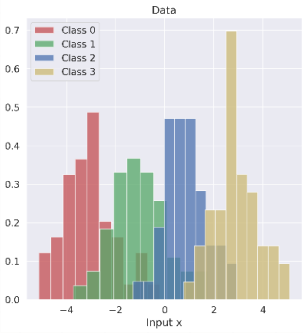
Computing logits:

$$f_n = W^T x_n + b = \begin{bmatrix} 1 & 0 & -1 \\ 0.5 & -0.5 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 0 \cdot 0 + (-1) \cdot 2 \\ 0.5 \cdot 1 + (-0.5) \cdot 0 + 0 \cdot 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0.5 \end{bmatrix}$$

taking a softmax: $e^{f_n} \approx [7.39, 0.37, 0.37]$, $\text{Sum} = 8.13$, $\hat{\pi}_n \approx [0.909, 0.046, 0.046]$

$\therefore P(y_n = 1 | x_n) \approx 0.91$ and the predicted class is 1

Log. likelihood of the Bayesian Linear softmax



Expected utility

y

	$\hat{y}$			
	1	0	0	0
	0	1	0	0
	0	0	1	0
	0	0	0	1

$U[i, j]$
true class
predicted class

100

10000

1036